

Full Analysis of ADA Architecture

Pump & Water Circuit, and Thermal Melting Performance

Nebico Oil Retention Project – CHL2-30 Pump Circuit and 10 kL
Palm Oil Heating Tanks (1D Radial Enthalpy Model)

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This supersedes my earlier report. The pump and water-circuit analysis is unchanged – what the CHL2-30 can really deliver, and the VFD settings that go with it. What's new is the tank thermal model: the old 2-zone (core/shell) model has been replaced with a proper 1D radial, 100-shell enthalpy model that also accounts for the steel tank wall, and it's changed the picture substantially – not because the hardware is any different, but because it reveals that *how* the system is operated (starting each batch cold versus carrying heat over from the last one) matters more than anything else. Nothing here proposes changing the pump, the coil or the heater – this is still about understanding and running the hardware we already have, with insulation as the one optional addition worth considering.

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1 Summary

This is the short version. The rest of the document backs it up in more detail.

The pump and water circuit – unchanged. The CHL2-30 pump still can't reach the old 4.04 m³/h target, and the same VFD settings apply: roughly 47 Hz for three tanks open, 50 Hz for two, and 44 Hz for one. Nothing in this section has changed since the last edition – see Section 2.

The tank model has been rebuilt, and the old numbers don't hold up. The previous 2-zone (core/shell) model was a reasonable first pass, but it split the oil 30/70 fairly arbitrarily. The coil actually sits at 80% of the tank's radius, not in the middle, so I rebuilt the model properly: 100 thin concentric shells from the centre of the tank out to the wall, tracking real phase-change physics shell by shell, including the thermal mass of the steel tank itself. Section 3 covers why, Section 4 covers how.

The single biggest finding: how you start the batch matters more than the weather. Starting a tank cold (at ambient temperature) means the coil can only deliver about 1.4 kW into solid oil – so little that the tank sits in a "dead time" of anywhere from 6 to 40 hours, depending on how cold it is outside, before any oil melts at all. **At Kathmandu winter temperatures, a cold-started tank produces zero usable liquid oil within a normal 24-hour working day.** But if the tank is never allowed to cool down between batches – carried over at around 30°C from the previous run – that dead time disappears completely. Liquid production starts within the first hour, and ambient temperature barely matters any more. See Section 5 and Section 6.

What that means for daily output. Cold-started, a single tank produces nothing useable in a day. Carried over, a single tank produces roughly 4,000–4,400 kg of liquid oil in 24 hours regardless of the weather, and three tanks run together in carryover mode produce something in the region of **12,000–12,800 kg/day** – several times over both the 2 t/day target and the 4 t/day satisfactory mark. See Section 7 for the full comparison.

The heater is still oversized – and the margin is bigger than we thought. During the dead time, the coil only needs about 1.4 kW out of the heater's 36 kW rating – under 4% duty. Once melting is properly under way, that rises to roughly 11–12.5 kW, or about a third of rated capacity. See Section 8.

Bottom line. The single most valuable change available to us costs nothing to install: **never let the tank fully cool down between batches.** That alone turns this from a system that can't hit target in winter into one that clears the satisfactory mark by 3× on any given day. Insulation remains worth adding on top of that – it stretches how long a tank stays in carryover condition and makes the outside weather close to irrelevant. See Section 9.

1.1 2-Tank Operation: Cold Start vs. Hot Start

Flow-rate note. These runs use 1.65 m³/h per tank (the 2-tank VFD setting), not the 1.17 m³/h used for the 3-tank figures elsewhere in this report. The 2-node comparison showed 1.65 m³/h running 3–8% ahead of 1.17 m³/h, so if anything the numbers below are slightly conservative.

Table 1: Total daily production, 2 tanks combined.

Start Condition	7°C (Winter)	18°C (Normal)	22°C (Best)
Cold Start (oil at ambient)	0 kg	4,749 kg	6,649 kg
Hot Start (30°C residual)	~7,950 kg	~8,495 kg	~8,680 kg

Table 2: Breakdown by tank.

Case	Tank 1 (24 h)	Tank 2	Total
Cold, 7°C	0 kg	0 kg	0 kg
Cold, 18°C	2,691 kg	2,058 kg	4,749 kg
Cold, 22°C	3,641 kg	3,008 kg	6,649 kg
Hot, 7°C	~4,190 kg	~3,760 kg	~7,950 kg
Hot, 18°C	4,405 kg	~4,090 kg	~8,495 kg
Hot, 22°C	4,405 kg	~4,275 kg	~8,680 kg

Table 3: Time to reach 4,000 kg combined liquid, 2 tanks.

Start Condition	8°C	18°C	22°C
Cold Start	~40 h	~20 h	~14 h
Hot Start	~12 h	~11.5 h	~11 h



2 The Pump and Piping Circuit

2.1 Layout

Water runs from a 1" supply line into a 1.5" main, then splits at a T-joint into the tank coils – Tank A on its own branch, and Tanks B, C and D sharing a further split downstream. Each coil is a spiral heat exchanger, and every coil is rated at about 7 metres of head loss when it's passing its design flow of roughly 1.35 m³/h. All the returns join back into a common 1.5" return line back to the pump. The whole circuit is CPVC pipe, carrying water between 40°C and 70°C.

2.2 Where the Original Target Went Wrong

The original design figure was 4.04 m³/h, based on wanting about 2.8 m/s velocity in the 1" branch pipe. Working through the head loss properly, though, showed two things. First, the 7 m figure is *per coil*, not for the whole bank – since the coils sit in parallel, the pump only has to overcome the head loss of *one* coil (plus the ordinary pipe friction), not all of them added together. Second, and more importantly: even accounting for that, our pump physically cannot reach 4.04 m³/h. That number isn't achievable with the CHL2-30 pump we have, and the figures below are what it can genuinely do instead.

2.3 What the Pump Can Actually Do

CHL2-30LSWSC, with a 0.37 kW (0.5 HP) motor. Reading off the pump's own performance curve: maximum flow is about 3.5 m³/h, maximum (shutoff) head is about 28 m, and the head available at 3.5 m³/h is about 12 m. Trying to push the pump past 3.5 m³/h risks overloading the motor – that figure is a hard ceiling for this hardware, not a target to design around.

2.4 Why the Number of Open Coils Changes Everything

This comes down to simple hydraulics. With all three tanks open, the flow splits three ways, so each coil only has to pass a third of the total – and because head loss rises with the *square* of flow through a given coil, spreading the flow out this way keeps total resistance low and lets the pump move more water overall. Close two of the tanks down to run just one, and all the flow has to go through a single coil – resistance shoots up, and the achievable flow drops a lot. This is just how parallel versus series resistance works; it's not something the VFD can get around, only something it has to be set correctly for.

2.5 VFD Settings by Number of Tanks Open

All the manual ball valves are assumed fully open throughout – flow control is handled entirely by the VFD frequency, not by throttling valves.

Tanks Open	VFD Frequency	Total Flow	Flow per Coil	Notes
3 Tanks	~47 Hz	3.5 m ³ /h	~1.17 m ³ /h	At 50 Hz the pump would try to push ~4.0 m ³ /h and overload the motor – 47 Hz holds it at the 3.5 m ³ /h ceiling.
2 Tanks	50 Hz	~3.3 m ³ /h	~1.65 m ³ /h	This is a natural balance point – at full 50 Hz the pump settles here on its own, safely under the motor limit.
1 Tank	44 Hz	2.0 m ³ /h	2.0 m ³ /h	A single coil creates a lot of backpressure; 44 Hz targets exactly 2.0 m ³ /h (1.0 m/s), keeping the motor at a comfortable ~0.22 kW draw.

2.6 Checking the Pipe Isn't Being Abused

CPVC loses pressure rating fast as it gets hotter – at 70°C its pressure rating drops to around 40% of what it is at room temperature – so it's worth checking velocity stays well inside the recommended 1.5 m/s ceiling at every flow setting, particularly at the 1" nozzle where velocity is highest.

Scenario	Velocity at 1" Nozzle	Verdict
3 Tanks (3.5 m ³ /h total)	~0.58 m/s	Safe
2 Tanks (3.3 m ³ /h total)	~0.82 m/s	Safe
1 Tank (2.0 m ³ /h total)	1.0 m/s	Safe, comfortably under 1.5 m/s

All three settings sit well inside the limit, so none of the VFD points above put any unusual strain on the pipework.

One thing worth flagging. The thermal simulations in Sections 5–7 were all run at a fixed 1.17 m³/h and 1.65 m³/h water flow – the 3-tanks-open per-coil rate above – rather than varying flow with the number of tanks actually running, the way the pump section does. I haven't re-run the thermal cases at 2.0 m³/h for this edition; the flow-rate sensitivity in the earlier report showed it's a second-order effect next to ambient temperature and, now, start condition, so I don't expect it changes the conclusions much – but it's not been explicitly checked at the other VFD point, and I'd want to before treating single-tank operation as fully characterised.

3 Why the Tank Thermal Model Was Rebuilt

The old 2-zone model (core near the coil, shell near the wall, split 30/70) was a reasonable first attempt, but rebuilding it properly meant going through the same evolution I went through the first time, one more step further.

3.1 Version 1 – One Lump of Oil (Rejected)

The simplest possible model treats the whole 9,200 kg of oil as one uniform temperature. It predicted the oil would reach 40°C in about 40 hours. That’s fundamentally misleading: solid palm oil has a thermal conductivity of about 0.18 W/mK, similar to wood – heat cannot spread through a solid block that fast. A single-temperature model assumes perfect mixing that solid oil simply cannot do.

3.2 Version 2 – Core and Shell (Partially Successful, Now Retired)

The version behind my previous report split the oil into a Core (30%, around the coil) and a Shell (70%, against the tank wall). This correctly revealed the “hot pocket” effect – the core would melt while the shell stayed cold – but the 30/70 split itself was arbitrary. The coil actually sits at a radius of about 0.98 m inside a 1.22 m-radius tank – 80% of the way out, not in the middle – so the real “shell” region is both inside *and* outside the coil helix at once. The old model was qualitatively right (hot pocket, cold shell) but quantitatively unreliable about exactly how much melts and when.

3.3 Version 3 – 1D Radial Enthalpy Model (Final, Used Throughout This Report)

The final model slices the tank into 100 thin concentric rings from the centre out to the wall, tracks each ring’s heat content (not just its temperature – more on why in Section 10) individually, and includes the 5 mm steel tank wall as its own thermal mass rather than ignoring it. This is what every result from Section 5 onward comes from.

Why this matters in practice. With the coil at 80% of the tank’s radius, there’s a large, nearly-insulated inner core (roughly two-thirds of the tank’s cross-section) that only ever warms up by slow conduction from the coil band – it never touches the wall, so it can’t lose heat to the outside either. That inner mass behaves like a thermal battery: slow to charge, but once warm, it holds its heat. The thin annulus between the coil and the wall, by contrast, is in direct contact with the cold steel and behaves like a leaky bucket – it’s what’s actually responsible for most of the dead-time problem in Section 5.

4 Results: Cold Start Operation

Cold start means the oil (and the steel tank) begins at ambient temperature – effectively, the tank has been sitting idle and lost all its retained heat before the batch starts. No insulation, 1.17 m³/h flow throughout.

4.1 Single Tank

Ambient	Time to 4 t	Final Liquid (200 h)	Dead Time
2°C	74.02 h	6,322 kg	~40 h
8°C	57.02 h	6,877 kg	~28 h
10°C	52.02 h	6,964 kg	~26 h
13°C	45.02 h	7,049 kg	~20 h
18°C	35.02 h	7,480 kg	~12 h
22°C	29.02 h	7,632 kg	~6 h

None of these fit inside a 24-hour working day. Even in the best case (22°C), the tank needs about 6 hours before any liquid appears at all, and 29 hours before it reaches 4 tonnes. A cold-started tank, run on its own, simply cannot deliver a same-day result at any Kathmandu ambient temperature.

4.2 The Dead Time Phenomenon

During the dead time, the coil is transferring heat into still-solid oil, where the oil-side heat transfer coefficient is only about 5 W/m²K – pure conduction, nothing else – so the coil can only actually deliver around **1.4 kW**, regardless of how much the 36 kW heater could otherwise put out. Once the band next to the coil finally reaches 31°C and starts to melt, the oil-side transfer coefficient jumps to 50+ W/m²K, and the whole system picks up sharply – a positive feedback loop, since faster melting means more liquid oil right at the coil, which transfers heat even better.

Ambient	Dead Time	Band Temp. at End	Coil Power at Start
2°C	~40 h	~29°C	1.4 kW
8°C	~28 h	~30°C	1.4 kW
13°C	~20 h	~31°C	1.4 kW
22°C	~6 h	~31°C	1.4 kW

The band temperature jump is sudden once it gets going – at 8°C ambient, for example, the band sits at only ~29.0°C at the 24-hour mark, but has jumped to ~55.6°C by the 36-hour mark. That’s the feedback loop in action, not a modelling artefact.

4.3 Three Tanks Combined

Running all three tanks at once, each at the same 1.17 m³/h, they all follow the same curve individually – the combined total reaches 4 tonnes once each tank has produced about 1,333 kg.

Ambient	Single-Tank Time	Combined 3-Tank Time	Speed-Up
2°C	74.0 h	~46 h	1.6×
8°C	57.0 h	~35 h	1.6×
10°C	52.0 h	~32 h	1.6×
13°C	45.0 h	~28 h	1.6×
18°C	35.0 h	~22 h	1.6×
22°C	29.0 h	~18 h	1.6×

Running three tanks doesn't triple output – it gives 1.6×, consistently.

The dead time itself doesn't shrink with more tanks running – all three sit through essentially the same dead time in parallel, since each tank's coil is fighting the same solid-oil bottleneck on its own. The 3× speed-up only shows up *after* the dead time has passed, so averaged over the whole run it comes out closer to 1.6×. More tanks helps, but it doesn't fix the underlying problem – that's still the cold start itself.

5 Results: Carryover Operation

Carryover means the tank starts the batch already warm – around 30°C, retained from the previous batch rather than allowed to go fully cold. Same 1.17 m³/h flow, no insulation, 24-hour run.

5.1 Single Tank

Ambient	Solid	Slurry	Liquid	Steel Temp. (end)	Avg. Liquid Temp.
2°C	3,819 kg	1,937 kg	4,039 kg	7.53°C	53.2°C
8°C	3,629 kg	1,945 kg	4,221 kg	13.09°C	52.9°C
10°C	3,629 kg	1,945 kg	4,221 kg	14.97°C	53.0°C
13°C	3,629 kg	1,945 kg	4,221 kg	17.76°C	53.1°C
18°C	3,437 kg	1,953 kg	4,405 kg	22.41°C	52.8°C
22°C	3,243 kg	2,147 kg	4,405 kg	26.11°C	53.0°C

Four things stand out here, compared with cold start:

- **No dead time.** Liquid production begins within the first hour, every time.
- **8, 10 and 13°C give identical results.** Once carried over, the oil's own thermal inertia swamps small differences in ambient temperature across that range.
- **Even the worst case (2°C) still produces 4,039 kg** – comparable to what cold-start achieves at 22°C, except in 24 hours rather than 29.
- **The 2 to 22°C ambient range changes output by less than 10%.** Carryover, not weather, is what's actually setting the outcome.

5.2 Three Tanks Combined

Assuming three tanks running staggered starts (roughly 24, 24, and 18.4 hours respectively, so the total lines up to a full day):

Tank	Duration	Liquid Production
Tank 1	24.0 h	4,221 kg
Tank 2	24.0 h	4,221 kg
Tank 3	~18.4 h	~3,800 kg (extrapolated)
Total	–	~12,242 kg/day

Time to reach 4 tonnes combined:

$$t = \frac{4,000}{12,242/24} \approx \mathbf{7.6 \text{ hours}}$$

That's the headline number: in carryover mode, three tanks clear the 4-tonne mark in well under a single work shift.

6 The 13°C Threshold

Cold-start results show a fairly sharp threshold around **13°C** ambient:

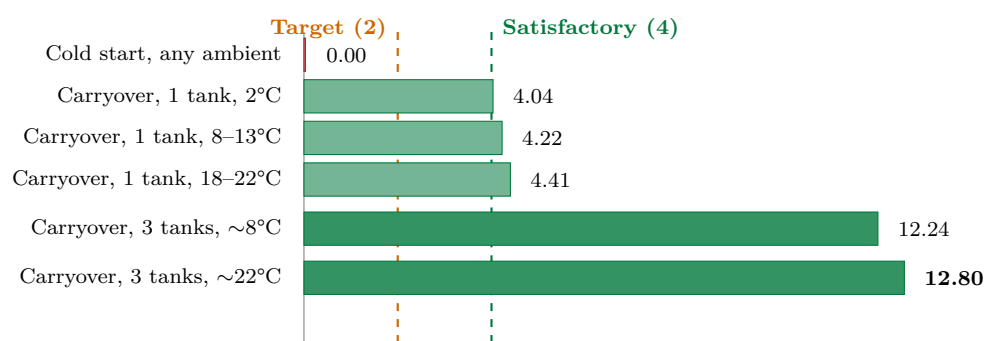
- **Below 13°C**, the tank is fighting a losing battle in cold start – the coil’s ~1.4 kW can’t keep up with how fast the uninsulated tank loses heat to the outside air, so the dead time stretches out badly (20 hours at 13°C, all the way to 40 hours at 2°C).
- **At 13°C**, melting finally begins after about 20 hours.
- **Above 13°C**, the picture improves quickly – dead time drops to single-digit hours by 22°C.

This threshold effectively disappears in carryover mode. Starting at 30°C puts the coil straight into the high-heat-transfer (melting) regime from the first minute, so the 13°C line – and ambient temperature generally – stops being the deciding factor. It only matters if the tank is ever allowed to go fully cold.

7 Daily Throughput Against Target

Our target was **2 t/day**, with **4 t/day** counted as a fully satisfactory result.

Mode	Scope	Output/Day	Rate	Status
Cold Start	Single tank, any ambient	0 kg	0.00 t/day	Below Target
Carryover	Single tank, 2°C	4,039 kg	4.04 t/day	Satisfactory
Carryover	Single tank, 8–13°C	4,221 kg	4.22 t/day	Satisfactory
Carryover	Single tank, 18–22°C	4,405 kg	4.41 t/day	Satisfactory
Carryover	3 tanks combined, ~8°C	~12,242 kg	12.24 t/day	3× over
Carryover	3 tanks combined, ~22°C	~12,800 kg	12.80 t/day	3× over



The gap between the top bar and the bottom two isn’t subtle – it’s the difference between a system that misses target every winter day and one that clears the satisfactory mark three times over, and the only thing that changed between them is whether the tank was allowed to go cold.

8 Heater Duty Cycle

The 36 kW heater’s spare capacity is even larger than the earlier report suggested, once the dead-time regime is accounted for separately from active melting.

Phase	Coil Power Draw	Duty Cycle Needed	Notes
Dead time (oil fully solid)	~1.4 kW	~3.9%	Barely any heat is actually reaching the oil during this phase – almost all the heater’s capacity is unused.
Active melting	~11–12.5 kW	~31–35%	Consistent with the earlier duty-cycle finding – the heater still never needs more than about a third of its rated output, even at the busiest point in the batch.

$$\text{Duty Cycle} = \frac{\text{Coil power draw}}{\text{Heater’s rated output (36 kW)}}$$

In plain terms: for most of a cold-started batch, the heater is doing almost nothing useful – which is really a symptom of the dead-time problem in Section 5, not evidence the heater is well matched to the job. In carryover mode, the system spends far more of its time in the “active melting” row above, using the heater’s real capacity properly rather than idling through a dead zone. Either way, nothing here calls for a different heater – it’s about how the existing one is controlled and, more importantly, when it actually gets useful work to do.

9 Solar Coverage

9.1 Basis

Solar generation (E_{solar}) is carried over unchanged from the FPC configuration used before – panel area, η_{FPC} , and $\Delta T = 70 - T_a$ don't depend on tank count, flow rate, or start condition, only on ambient temperature – and, as before, I'm treating those figures as a **conservative estimate**. What's changed is the demand side: the old 2-node model's energy table is retired along with the rest of that model, so daily energy demand has to be rebuilt from what the 1D radial model actually gives us – coil power by phase, not a direct kWh/day figure.

This is a derived estimate, not a simulated result. The new model reports coil power at two points – ~ 1.4 kW while the oil is still solid (dead time), ~ 11 – 12.5 kW once melting is properly under way – but not a continuous power curve. To get a daily energy figure, I split each 24-hour day into dead-time hours (from Section 5's dead-time table, capped at 24 h) and melting-phase hours, using the **midpoint of the melting range, 11.75 kW**, held flat for the whole melting portion of the day. Real melting-phase power actually tapers as the band warms toward 70°C over the day (it's already at a mass-weighted $\sim 53^\circ\text{C}$ by the end of a carryover run), so holding it flat at the midpoint *overstates* energy demand – which means the coverage percentages below are, if anything, conservative in the same direction as the solar figures themselves.

9.2 Coverage by Case

Ambient	Mode	Dead (h)	Melt (h)	Est. Energy (kWh/day)	Solar Gen. (kWh/day)	Coverage
2°C	Cold Start	24.0	0.0	33.6	22.35	66.52%
2°C	Carryover	0.0	24.0	282.0	22.35	7.93%
18°C	Cold Start	12.0	12.0	157.8	29.82	18.90%
18°C	Carryover	0.0	24.0	282.0	29.82	10.57%
22°C	Cold Start	6.0	18.0	219.9	29.78	13.54%
22°C	Carryover	0.0	24.0	282.0	29.78	10.56%

9.3 Interpretation

The 66.52% figure at 2°C cold start is misleading, not good news. At 2°C , cold-start dead time runs ~ 40 hours – longer than the whole 24-hour window – so the entire day sits at ~ 1.4 kW, and the tank produces **zero liquid oil** (Section 7). The high coverage number here isn't solar doing well; it's the same paradox flagged in the earlier report – when almost no real heating is happening, a small solar contribution covers a large share of a very small, mostly-wasted demand. This number shouldn't be read as "solar performs best in winter."

Set that case aside, and the more honest picture is in the carryover rows: coverage sits in a narrow **7.93–10.57%** band across all three ambients, because carryover keeps the coil in the melting-power regime for essentially the whole day regardless of weather – this

is what solar actually buys us when the system is doing real work. The cold-start rows at 18°C and 22°C (18.90% and 13.54%) sit in between: real melting is happening for part of the day, but the long dead-time tail at the start still drags total demand down relative to carryover, inflating coverage somewhat by the same mechanism as the 2°C case, just less severely.

In absolute terms, the solar figures themselves don't change with operating mode – **22.35 kWh/day at 2°C, 29.82 kWh/day at 18°C, 29.78 kWh/day at 22°C** – it's only the demand side, and therefore the coverage percentage, that moves. Carried over, that's a consistent offset of roughly a tenth of daily electric demand; cold-started, the same absolute saving looks larger only because the system is producing little or nothing to offset it against.

10 What This Means for Running the System

To be clear about scope: as before, this is about how to *operate* the pump, coils and heater we already have – none of it calls for a bigger pump, a different coil, or a resized heater.

- **Build the carryover threshold directly into the ADA program.** Rather than leaving "never let the tank go cold" as a manual habit, I'll add a threshold-triggered reheat routine to the ladder logic itself – whenever the system is otherwise free, it keeps the tank circulating and warm instead of sitting idle. The open question is exactly which threshold to use: the RTD sensor sits closer to the tank wall than to the core, so what it actually reads and what the simulation calls the band or core temperature aren't the same thing. I want to do more work to find the threshold that's genuinely efficient against what the sensor can really see, rather than lifting a number straight from the simulation and assuming it translates directly.
- **Insulation doesn't clear the bar – but a slow impeller stays on the table.** Adding insulation isn't a call I can make anyway, and my own numbers don't build a strong case for it regardless: 50 mm of mineral wool only bought about 6% more production at cold start, which doesn't justify the cost. What I'd keep open instead is a small impeller – even a slow one – to agitate the solid mass. The solid oil layer at cold start is the real bottleneck every simulation keeps pointing at, and a slow stir there looks like it could move the needle far more than insulation ever would. Even a manual agitation from the top of manhole via mixing instrument is open to operation.
- **Keep the VFD settings from Section 2, with a caveat noted below for whoever commissions them.** Roughly 47 Hz for three tanks, 50 Hz for two, 44 Hz for one – these haven't changed, but see the note on how they were calculated.
- **The heater's spare capacity is a safety question as much as a sizing one.** With only a 400 L loop, the 36 kW heater could boil that water in under an hour at full throttle if nothing were managing it. The program already accounts for this – T_{target} adjusts the water setpoint and latches/de-latches heater stages for a smoother transition – but see the safety note below on where that protection can still be bypassed.

Note for whoever handles VFD commissioning. I may be away at school when the VFD is actually installed and tuned, so I want this on record: the pipe resistance network I used to calculate the 44/47/50 Hz figures was only half-modelled at the time – the remaining run was estimated rather than measured, since the piping wasn’t fully laid out yet. I expect the real optimum frequencies to land close to these figures, and the shift to be negligible, but they should be verified on site rather than taken as final – worth expecting a small adjustment rather than being surprised by one.

Safety note on the heater/manual-mode interaction. The T_{target} staging logic that keeps the heater from overshooting is theoretically sound but not yet practically tested against the real system. More importantly, it only governs automatic mode – in manual mode, where the operator has full control, that protection can be bypassed entirely. I’m flagging this now, before it becomes a live problem rather than a foreseen one: a 36 kW heater on a 400 L loop leaves very little margin for operator error.

11 Assumptions and Limitations

Worth being upfront about what this model does and doesn’t capture, since every number above comes from it.

11.1 What’s Assumed

Assumption	Why
1D radial symmetry	The coil is a helix; heat spreads outward/inward radially as a reasonable approximation
No vertical stratification	Tank is treated as uniform top to bottom
Constant ambient temperature	Fixed for the duration of each scenario, no day/night cycling
Constant water inlet (70°C)	Assumes the 36 kW heater holds setpoint throughout
Effective conductivity $\times 5$ in liquid regions	Stands in for natural convection mixing that a 1D model can’t otherwise capture
Perfect contact at the oil–wall interface	No contact resistance modelled
No water-side fouling	$R_{f,in} = 0$
50/50 heat split at the coil	Matches the coil’s own geometry – inner and outer surfaces see equal flux
No agitation	No forced mixing anywhere in the tank
Fixed water flow (1.17 m ³ /h)	Not varied with tank count in this edition – see the note in Section 2.7

11.2 What It Doesn't Capture

1. 3D natural convection currents in the liquid oil.
2. Vertical stratification (hot oil rising, cold oil sinking).
3. The true angular asymmetry of the helical coil.
4. Slurry pumpability – the model classifies slurry by mass and temperature, but doesn't predict whether it can actually be pumped.
5. Diurnal ambient swings – each scenario uses one fixed temperature for its whole run.
6. The 400 L water/heater loop's own control dynamics.
7. Heat loss through flanges, supports and instrumentation ports.
8. Contact resistance at the oil-steel interface.

None of these change the headline conclusion – carryover beats cold start by a wide enough margin that model refinement at this level of detail wouldn't close the gap – but they're the right caveats to carry forward into any decision that leans on the exact numbers rather than the overall pattern.

12 Details on the Modelling Method (For those who interested)

Everything above stands on its own – this section is for anyone who wants to see how the 1D radial model in Section 3 and Section 4 actually works underneath. It's not required reading to use the results.

12.1 The Radial Grid

The tank is discretised into $N = 100$ concentric cylindrical shells of equal thickness:

$$\Delta r = \frac{R_{\text{tank}}}{N} = \frac{1.22}{100} = 0.0122 \text{ m}$$

Shell i sits at radius $r_i = (i + 0.5)\Delta r$ with mass

$$m_i = \rho_{\text{oil}} \cdot \pi [(r_{i+1})^2 - (r_i)^2] H_{\text{tank}}$$

The coil sits at shell index $i_{\text{coil}} = \lfloor 0.98/0.0122 \rfloor = 80$ – 80% of the way out from the centre. A hundred shells was chosen as a balance point: fewer (say 20) blur the melt front badly; many more (say 500) would need a much smaller time step to stay numerically stable for not much gain in accuracy.

12.2 Tracking Enthalpy Instead of Temperature

When palm oil melts, it absorbs 65 kJ/kg of latent heat with *no* temperature change. Tracking temperature directly would mean artificially inflating C_p during melting – a hack that gets numerically unstable fast across 100 coupled shells. Instead, the model tracks specific enthalpy H (J/kg), which rises smoothly regardless of phase:

$$H(T) = \begin{cases} C_{p,\text{solid}} \cdot T & T < 31^\circ\text{C (solid)} \\ C_{p,\text{solid}} \cdot 31 + L \cdot \frac{T - 31}{10} & 31 \leq T \leq 41^\circ\text{C (slurry)} \\ C_{p,\text{solid}} \cdot 31 + L + C_{p,\text{liquid}}(T - 41) & T > 41^\circ\text{C (liquid)} \end{cases}$$

and the reverse mapping (used every step, to turn the updated enthalpy back into a temperature) simply inverts each branch. The local melt fraction then falls straight out of temperature:

$$f_{\text{melt}}(T) = \begin{cases} 0 & T < 31^\circ\text{C} \\ (T - 31)/10 & 31 \leq T \leq 41^\circ\text{C} \\ 1 & T > 41^\circ\text{C} \end{cases}$$

12.3 Coil Heat Injection

The coil is treated as a heat exchanger between the water inside it and the oil band immediately around it, using the standard NTU-effectiveness method:

$$Q_{\text{coil}} = \varepsilon \dot{m}_{\text{water}} C_{p,\text{water}} (T_{w,\text{in}} - T_{\text{band}}), \quad \varepsilon = 1 - e^{-NTU}, \quad NTU = \frac{UA_{\text{coil}}}{\dot{m}_{\text{water}} C_{p,\text{water}}}$$

with $A_{\text{coil}} = \pi d_{\text{out}} L_{\text{coil}}$. The overall U comes from the usual series resistance stack:

$$\frac{1}{U} = \frac{1}{h_{\text{in}}} + \frac{d_{\text{out}} \ln(d_{\text{out}}/d_{\text{in}})}{2k_{ss}} + R_{f,\text{out}} + \frac{1}{h_{\text{out}}}$$

Water-side h_{in} comes from a modified Gnielinski correlation (turbulent flow, with a curvature correction for the coil being helical rather than straight); oil-side h_{out} is a flat $\sim 5 \text{ W/m}^2\text{K}$ when the band is solid (pure conduction), or the Churchill–Chu natural-convection correlation, using the Andrade viscosity equation, once the band is liquid.

The 50/50 split. Heat from the coil is injected equally into the two shells immediately on either side of it (79 and 81) – physically justified because the coil is a thin pipe, so the inward-facing and outward-facing heat flux at its surface are geometrically equal:

$$H_{i_{\text{coil}}-1} += \frac{0.5 Q_{\text{coil}} \Delta t}{m_{i_{\text{coil}}-1}}, \quad H_{i_{\text{coil}}+1} += \frac{0.5 Q_{\text{coil}} \Delta t}{m_{i_{\text{coil}}+1}}$$

This is a strict rule in the model, not a tunable parameter. If the band reaches 70°C (the water inlet temperature), Q_{coil} is forced to zero – heat can’t flow from 70°C water into already- 70°C oil.

12.4 Radial Conduction Between Shells

Between every adjacent pair of shells, plain Fourier conduction applies:

$$Q_{\text{cond},i} = \frac{k_{\text{eff}} A_i (T_i - T_{i+1})}{\Delta r}, \quad A_i = 2\pi r_i H_{\text{tank}}$$

where k_{eff} is the solid oil conductivity (0.180 W/mK) below 31°C , a linear blend through the melting band, and **5 times** the liquid conductivity ($0.170 \text{ W/mK} \rightarrow 0.85 \text{ W/mK}$) above 41°C – the $\times 5$ factor is a standard heuristic for approximating the mixing effect of natural convection currents that a strictly 1D conduction model can’t otherwise represent, calibrated so melt behaviour matches expectations for an unstirred tank rather than derived from first principles. The centre of the tank ($r = 0$) is treated as adiabatic by symmetry – no heat crosses it – and the outermost oil shell hands its heat to the steel wall node across a half-shell gap.

12.5 The Steel Wall

The tank’s 5 mm mild-steel wall (about 690 kg) is carried as its own thermal node rather than ignored, since at the start of any cold batch it’s sitting at ambient temperature too and would otherwise make the model over-predict how fast the oil heats:

$$M_{\text{steel}} C_{p,\text{steel}} \frac{dT_{\text{steel}}}{dt} = Q_{\text{oil} \rightarrow \text{steel}} - U_{\text{tank}} A_{\text{tank}} (T_{\text{steel}} - T_{\text{amb}})$$

U_{tank} is a settable parameter – $10 \text{ W/m}^2\text{K}$ uninsulated (used throughout this report), or about $0.7 \text{ W/m}^2\text{K}$ with 50 mm mineral wool, which is what makes insulation such an effective lever on top of carryover operation.

12.6 Stepping Forward in Time

Every 60-second step, the model: computes Q_{coil} from the current band temperature; applies the 50/50 split; sweeps through all 100 shell-to-shell conduction pairs; computes oil-to-steel conduction and steel-to-air loss; updates every shell’s enthalpy; then converts each back to temperature. This is plain explicit forward Euler –

$$H_i^{n+1} = H_i^n + \frac{\sum Q_{\text{in}} - \sum Q_{\text{out}}}{m_i} \Delta t$$

– chosen for being simple, fast (100 shells $\times$ 60 s $\times$ 48 h is a few hundred thousand iterations, seconds on any laptop), and transparent about how errors propagate. The 60-second step comfortably satisfies the standard conduction stability limit, $\Delta t < \rho C_p \Delta r^2 / 2k \approx 760$ s for these shells, so there’s real margin before the scheme would need a shorter step.

12.7 Physical Constants Used

Constant	Value
Oil density, solid	920 kg/m ³
Oil density, liquid	$902.5 - 0.53T$ kg/m ³
C_p , solid oil	1,600 J/kgK
C_p , liquid oil	$1750 + 2.9T$ J/kgK
k , solid oil	0.180 W/mK
k , liquid oil	$0.174 - 0.00008T$ W/mK
Latent heat of fusion	65,000 J/kg
Melting range	31–41°C
Oil viscosity (Andrade)	$1.85 \times 10^{-6} e^{3080/(T+273)}$ Pa·s
Water C_p	4,185 J/kgK
Water k	0.654 W/mK (at $\sim 50^\circ\text{C}$)
Steel k	16 W/mK (SS304)
Steel ρC_p	7850×500 J/m ³ K

On the oil mass figure. This model computes the oil mass directly from tank geometry – 9,795 kg – rather than the rounded 9,200 kg used as a starting assumption in the earlier 2-zone model. The small difference doesn’t change any conclusion above; it’s simply a more precise figure now that it falls out of the geometry itself rather than being assumed up front.